HW 5

October 15, 2024

1. Calculate the derivative.

(1) $y = \left(x^2 + x^3\right)^4$	(2) $y = \frac{x^2 - x + 2}{\sqrt{x}}$	(3) $y = x^2 \sin \pi x$
(4) $y = \ln(x \ln x)$	(5) $y = \sqrt{x} \cos \sqrt{x}$	(6) $y = \frac{e^{1/x}}{x^2}$
(7) $y = \tan\left(\frac{t}{1+t^2}\right)$	(8)$y = 3^{x \ln x}$	(9) $y = \left(1 - x^{-1}\right)^{-1}$
(10)$y = \log_5(1 + 2x)$	(11)$y = \cos\left(e^{\sqrt{\tan 3x}}\right)$	(12) $y = \sin^2(\cos\sqrt{\sin \pi x})$

2. If $g(x) = x/e^x$, find $g^{(n)}(x)$.

3. Use the Reciprocal Rule and Product Rule to prove the Quotient Rule.

4. (a) Evaluate $\lim_{x \to \infty} x \sin \frac{1}{x}$.
(b) Evaluate $\lim_{x \to 0} x \sin \frac{1}{x}$.

5. Find the limit.

(1) $\lim_{x \to 0} \frac{\sin 5x}{3x}$	(2) $\lim_{t \to 0} \frac{\tan 6t}{\sin 2t}$	(3) $\lim_{x \to 0} \frac{\sin 3x}{5x^3 - 4x}$
(4) $\lim_{\theta \to 0} \frac{\sin \theta}{\theta + \tan \theta}$	(5)$\lim_{x \to 0} \csc x \sin(\sin x)$	(6) $\lim_{x \to 1} \frac{\sin(x-1)}{x^2 + x - 2}$

6. Suppose f is differentiable on $\mathbb{R}$. Let $F(x) = f(e^x)$ and $G(x) = e^{f(x)}$. Find expressions for (a) $F'(x)$ and (b) $G'(x)$.

7. If $y = f(u)$ and $u = g(x)$, where f and g are twice differentiable functions, show that

$$\frac{d^2y}{dx^2} = \frac{d^2y}{du^2}\left(\frac{du}{dx}\right)^2 + \frac{dy}{du}\frac{d^2u}{dx^2}.$$

8. Find the linearization $L(x)$ of $f(x) = \sqrt{x}$ at $x = 4$.

9. Use a linear approximation (or differentials) to estimate $e^{0.1}$.

Bonus

1. Evaluate $\lim_{x \to \pi} \frac{e^{\sin x} - 1}{x - \pi}$.

2. If f and g are differentiable functions with $f(0) = g(0) = 0$ and $g'(0) \neq 0$, show that

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = \frac{f'(0)}{g'(0)}$$